

Shiv Nadar University Chennai

CS3807 - Deep Learning Laboratory

Experiment 5

Comprehensive Study of CNN Training, Regularization, Optimization, Hyperparameter Tuning, Transfer Learning and Cross-Validation

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester	V
Subject Code & Name	CS3807 - Deep Learning Laboratory	AY	2026-27
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1 Objective

To systematically study the effect of weight initialization, regularization, optimization algorithms, CNN hyperparameters, transfer learning, fine-tuning and cross-validation on image classification performance using one MobileNetV2 architecture and the Oxford-IIIT Pet dataset.

2 Background Theory

A CNN learns local image patterns with shared convolution kernels. MobileNetV2 reduces computation with depthwise separable convolutions, inverted residual blocks, linear bottlenecks, Batch Normalization and ReLU6 activations. Global average pooling converts the final feature maps into a compact feature vector before the classifier.

The classifier used in the experiments was $1280 \rightarrow 128 \rightarrow 37$, with ReLU in the hidden layer and Softmax at the output. Cross-entropy loss was used for the 37-breed prediction task.

Component	Role
Depthwise convolution	Applies one spatial filter per input channel.
Pointwise convolution	Uses 1×1 filters to mix channels.
Inverted residual	Expands channels, applies depthwise convolution, then projects back.
Linear bottleneck	Uses a linear projection to preserve information in a narrow layer.
Global average pooling	Averages each final feature map into one value.
Transfer learning	Reuses ImageNet features for the pet-breed task.

3 Dataset and Experimental Setup

The Oxford-IIIT Pet Dataset contains RGB images from 37 cat and dog breeds. Every image was resized to $224 \times 224 \times 3$ and normalized with ImageNet mean and standard deviation. The official trainval split was divided into training and validation subsets with stratification. The official test split was kept independent and was not used during hyperparameter selection.

Item	Value
Training images	2,944
Validation images	736
Testing images	3,669
Image size	$224 \times 224 \times 3$
Classes	37 breeds
Device	mps
Split seed	24110085

One Oxford-IIIT Pet image from each breed

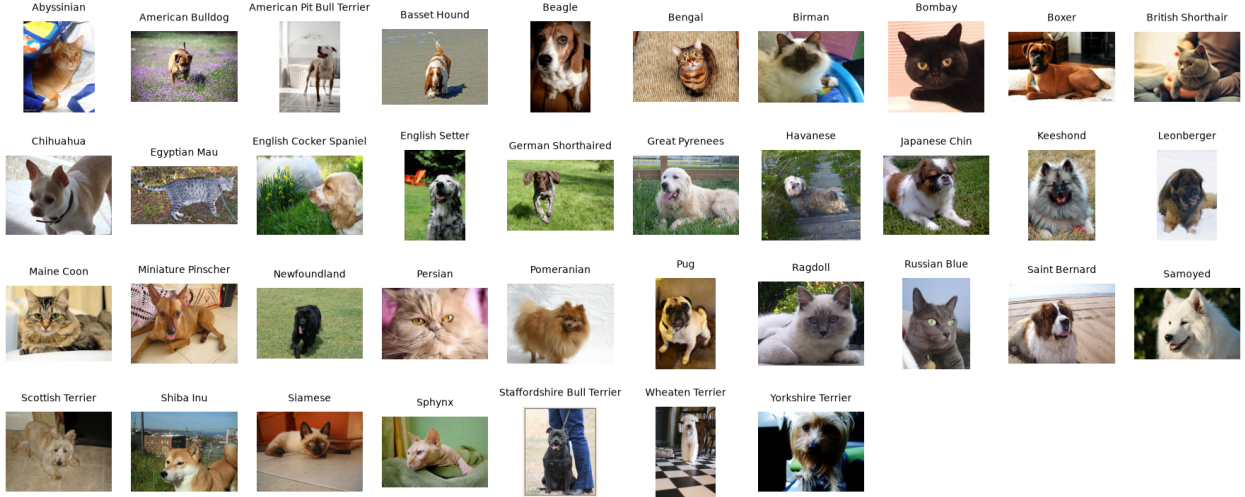


Figure 1: One representative Oxford-IIIT Pet image from each breed.

Inference: It was observed that all 37 breeds were represented and several visually similar breeds were difficult to distinguish.

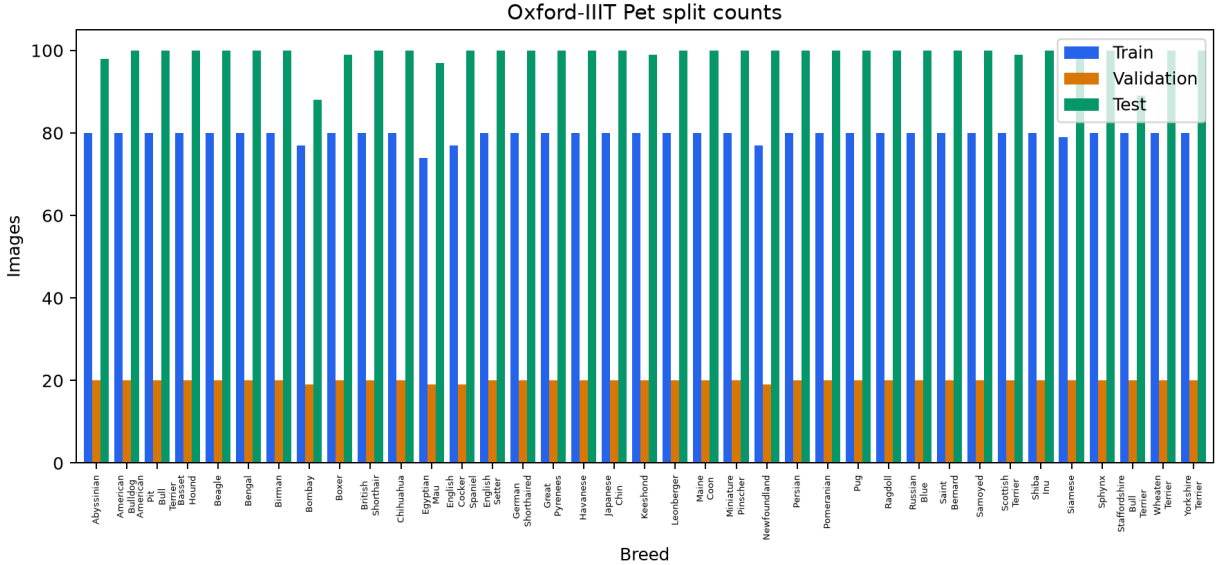


Figure 2: Train, validation and test counts by breed.

Inference: It was observed that the stratified split kept the training and validation class counts balanced while the test set remained separate.

The two dataset views were kept separate: transformed images were used for MobileNetV2 fine-tuning, while the same frozen pretrained base was used to cache 1280-dimensional global-average-pooled features for the controlled head comparisons and cross-validation.

4 Experimental Procedure

1. Load Oxford-IIIT Pet, verify dimensions and display one sample from each breed.
2. Load ImageNet-pretrained MobileNetV2 and replace its classifier with a 128-unit ReLU layer and a 37-class output layer.
3. Compare zero, random, Xavier/Glorot and He initialization using training-loss and validation-accuracy curves.
4. Compare no regularization, L2 regularization, Dropout and Batch Normalization using accuracy and loss curves.
5. Calculate the Batch Normalization numerical example and compare validation accuracy with and without BN.
6. Compare SGD, Momentum, RMSProp and Adam using loss, validation accuracy, convergence epoch and time.
7. Vary learning rate, batch size and Dropout one at a time from the reference setting; record the required fine-tuning settings.

8. Compare frozen feature extraction with partial fine-tuning of the upper MobileNetV2 blocks while Batch Normalization statistics remain frozen.
9. Use 5-fold cross-validation on training data to compare four promising configurations and two additional combinations.
10. Retrain the selected configuration and evaluate the independent test set using accuracy, precision, recall, F1-score, confusion matrix and the classification report.

The convergence epoch in the tables was defined as the first epoch whose validation accuracy was within 1 percentage point of that run’s best validation accuracy.

5 Numerical Examples and Dimension Calculations

5.1 Batch Normalization Example

For $x = [2, 4, 6, 8]$, the batch mean was

$$\mu_B = \frac{2 + 4 + 6 + 8}{4} = 5.$$

The population variance was

$$\sigma_B^2 = \frac{(2 - 5)^2 + (4 - 5)^2 + (6 - 5)^2 + (8 - 5)^2}{4} = 5,$$

so $\sqrt{5} = 2.236$. With $\gamma = 1$, $\beta = 0$ and negligible ϵ ,

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} = [-1.342, -0.447, 0.447, 1.342].$$

Kernel / setting for $N = 32$	Output size
3×3 , valid, stride 1	30×30
5×5 , valid, stride 1	28×28
7×7 , valid, stride 1	26×26
3×3 , same, stride 1	32×32
3×3 , valid, stride 2	15×15
3×3 , same, stride 2	16×16

The convolution output relation used was $O = \lfloor \frac{N+2P-K}{S} \rfloor + 1$.

6 Weight Initialization

Method	Best val. accuracy	Final loss	Epoch	Time (s)
Zero	2.72%	3.6113	1	0.61
Random	90.76%	0.0444	3	0.58
Xavier/Glorot	91.71%	0.0292	6	0.58
He	91.44%	0.0266	7	0.57

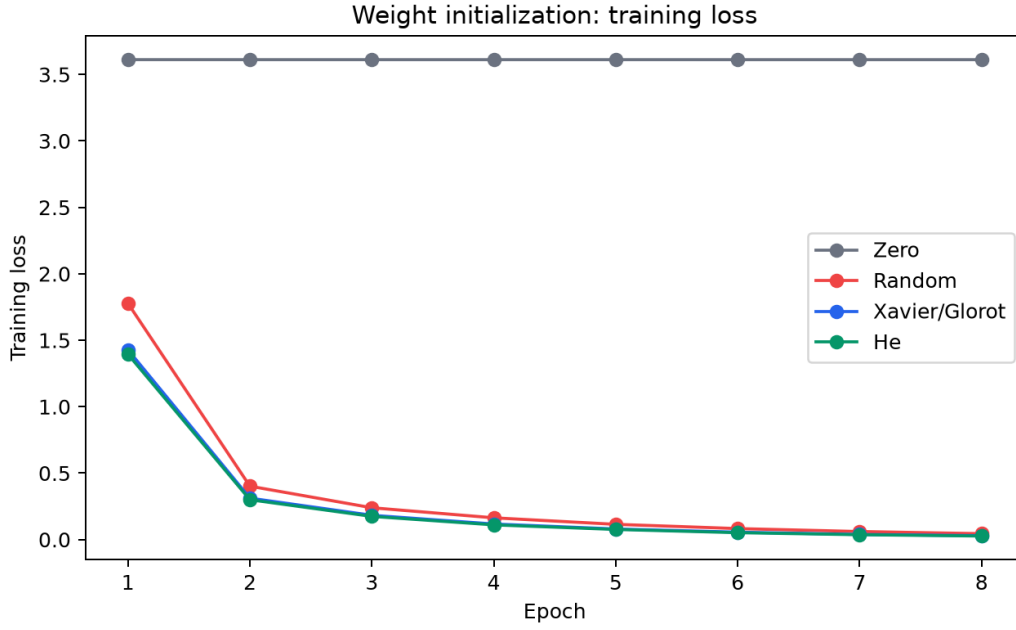


Figure 3: Training loss versus epoch for each initialization method.

Inference: It was observed that Xavier/Glorot and He reduced loss rapidly, whereas zero initialization remained almost unchanged.

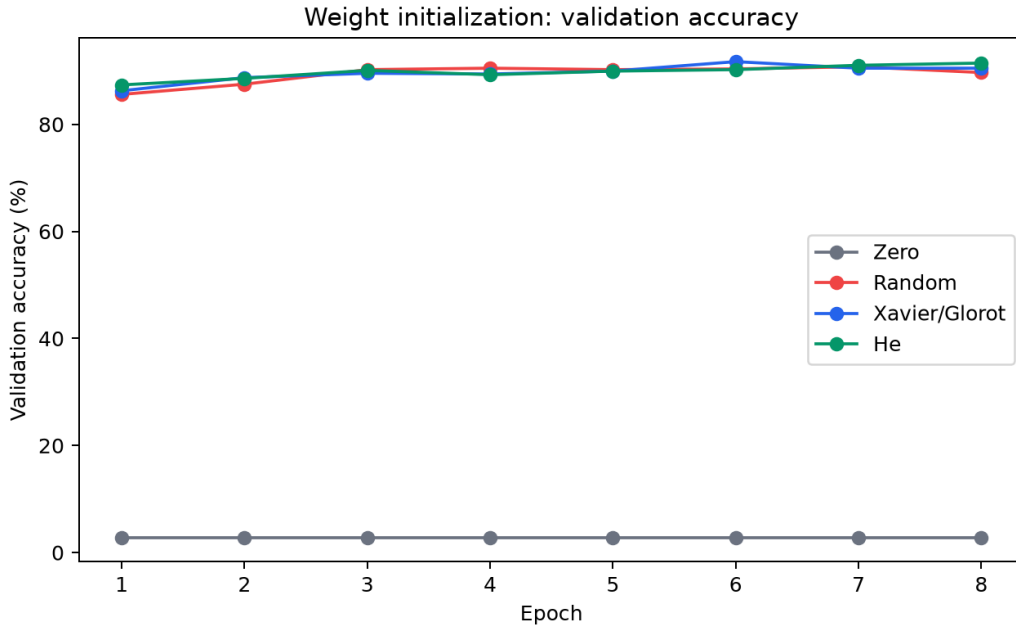


Figure 4: Validation accuracy versus epoch for each initialization method.

Inference: It was observed that the non-zero initializations reached high validation accuracy, while zero initialization stayed near chance level.

7 Regularization and Overfitting

Setting	Best val. accuracy	Train acc. final	Val acc. final	Val loss final
No regularization	90.76%	99.97%	90.62%	0.2816
L2	91.30%	100.00%	91.30%	0.2774
Dropout	91.30%	95.48%	91.17%	0.2924
Batch Normalization	91.30%	99.97%	91.30%	0.2804

Regularization: training and validation accuracy

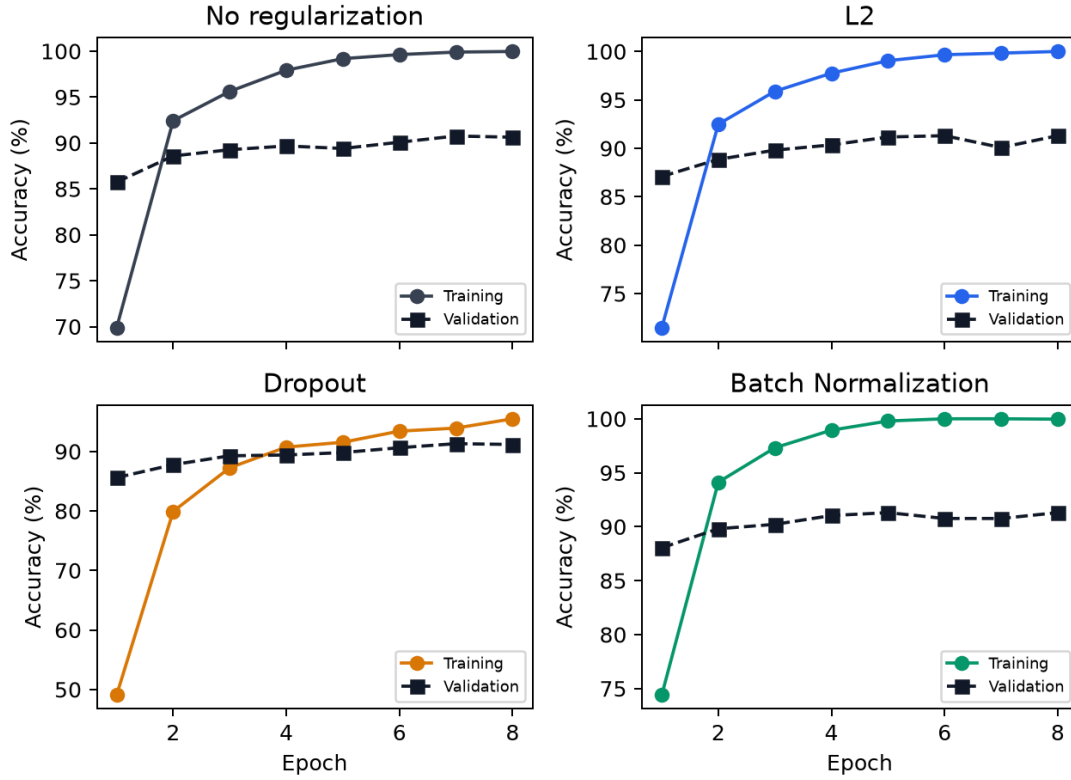


Figure 5: Training and validation accuracy for the four regularization choices.

Inference: It was observed that regularization reduced the train-validation gap and produced steadier validation behaviour.

Regularization: training and validation loss

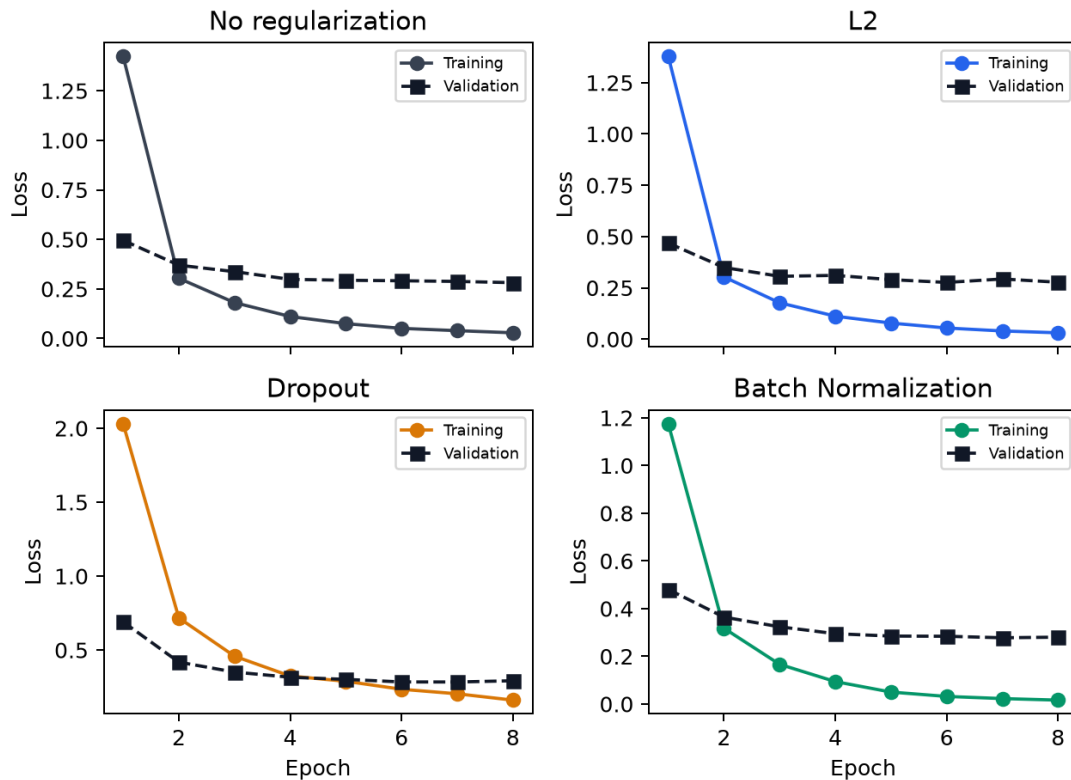


Figure 6: Training and validation loss for the four regularization choices.

Inference: It was observed that validation loss levelled off before training loss, indicating the beginning of overfitting.

8 Batch Normalization

The numerical result above was checked directly in the notebook. In the model study, the BN variant added BatchNorm1d after the first classifier layer; the pretrained MobileNetV2 BatchNorm2d layers were kept in evaluation mode during feature extraction and fine-tuning.

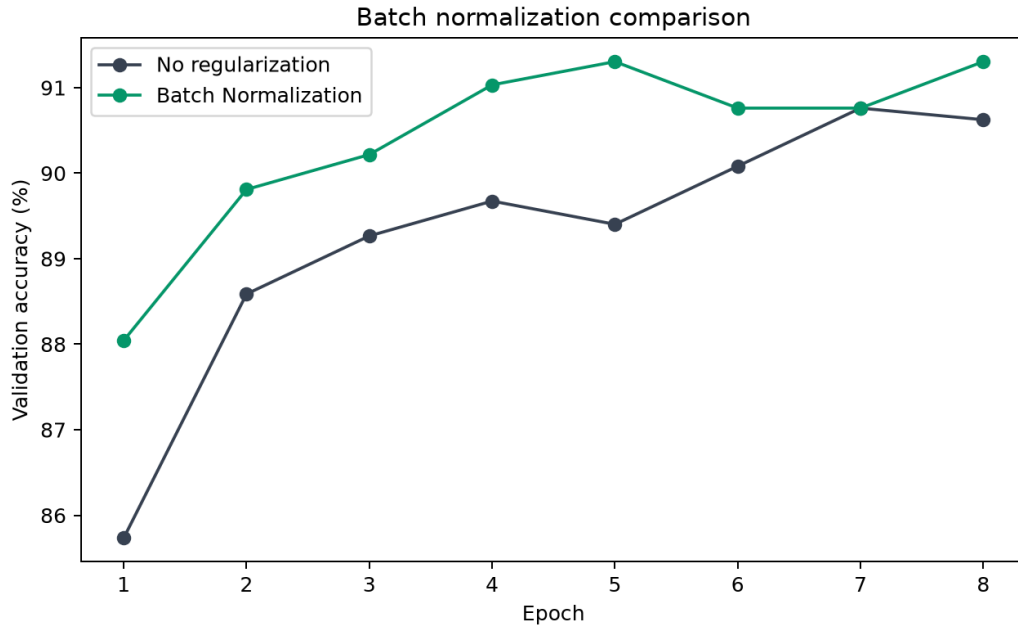


Figure 7: Validation accuracy with and without Batch Normalization.

Inference: It was observed that Batch Normalization produced a smoother validation curve and competitive final accuracy.

9 Optimization Algorithms

Optimizer	Final loss	Best val. accuracy	Epoch to converge	Time (s)
SGD	3.3256	23.23%	8	0.34
Momentum	0.8627	86.14%	8	0.43
RMSProp	0.0154	90.62%	4	0.50
Adam	0.0300	90.49%	3	0.59

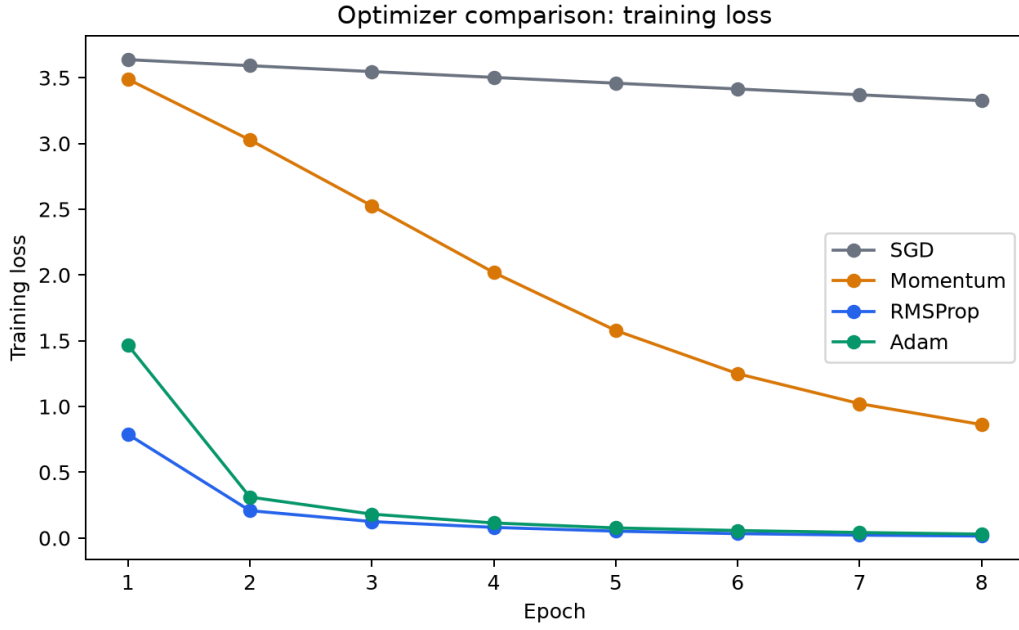


Figure 8: Training loss for SGD, Momentum, RMSProp and Adam.

Inference: It was observed that Adam and RMSProp reduced loss fastest, while plain SGD converged much more slowly.

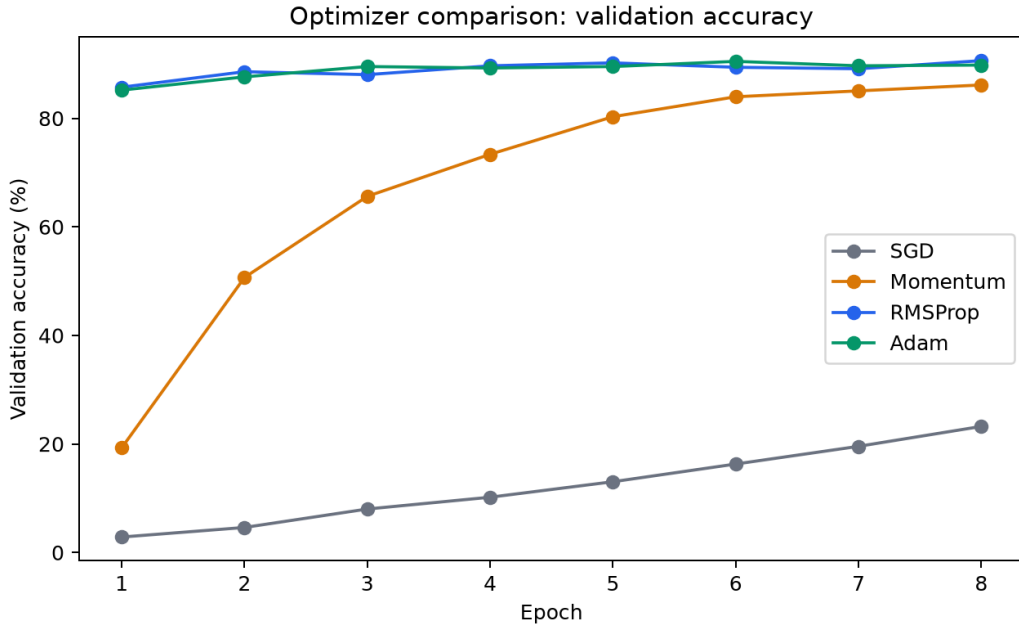


Figure 9: Validation accuracy for the four optimizers.

Inference: It was observed that the adaptive optimizers reached strong validation accuracy in fewer epochs than SGD and Momentum.

10 CNN Hyperparameter Tuning

The controlled study changed one value at a time from the reference setting: He initialization, Adam, learning rate 0.001, batch size 32, 128 hidden units, no Dropout and a frozen base. The independent test set was not used for these choices.

Study	Value	Batch	Dropout	Optimizer	Val. accuracy	Time (s)
Learning rate	0.001	32	0	Adam	91.17%	0.58
Learning rate	0.0001	32	0	Adam	88.04%	0.60
Batch size	16	16	0	Adam	91.44%	1.00
Batch size	32	32	0	Adam	91.17%	0.58
Batch size	64	64	0	Adam	91.58%	0.39
Dropout	0	32	0	Adam	91.17%	0.58
Dropout	0.25	32	0.25	Adam	91.17%	0.63
Dropout	0.5	32	0.5	Adam	91.30%	0.60

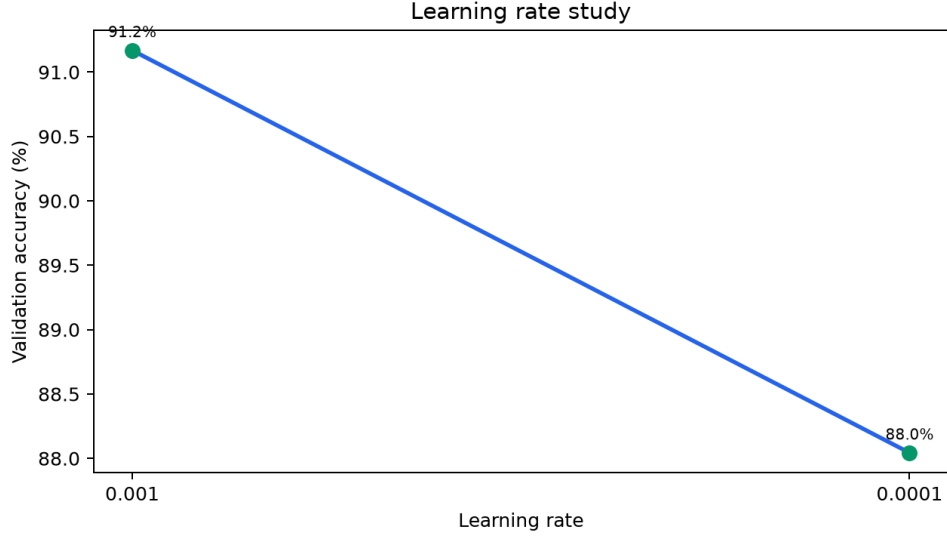


Figure 10: Learning rate versus validation accuracy.

Inference: It was observed that the 0.001 learning rate achieved higher validation accuracy within the fixed epoch budget.

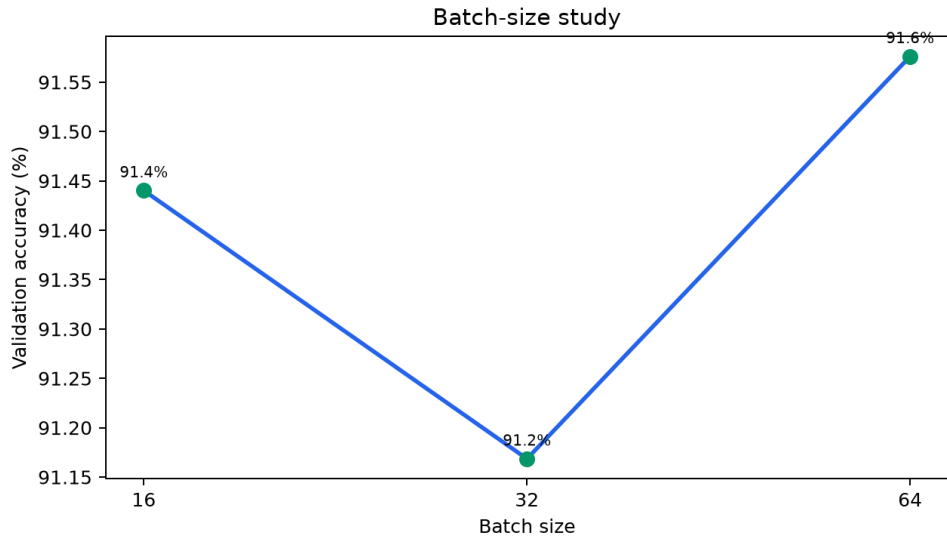


Figure 11: Batch size versus validation accuracy.

Inference: It was observed that the batch-size results were close, with batch size 64 giving the highest recorded validation accuracy.

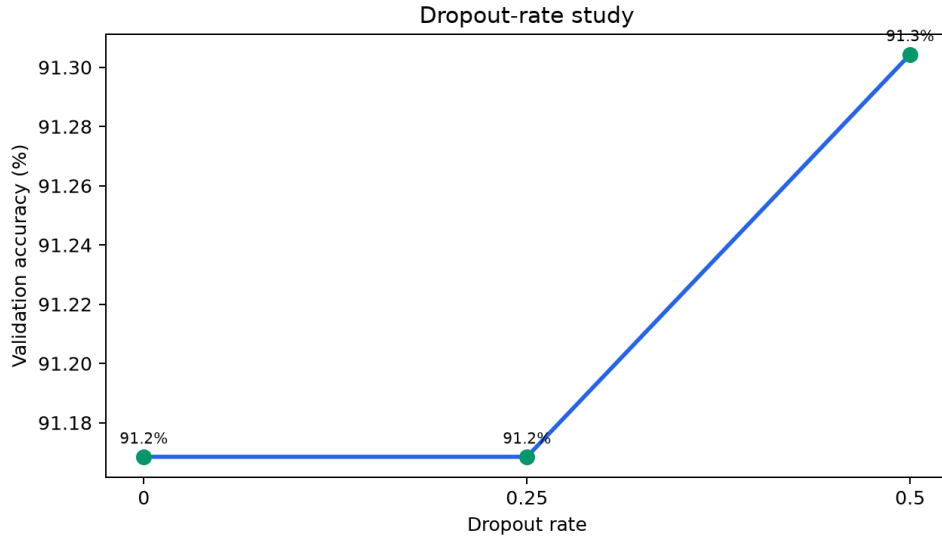


Figure 12: Dropout rate versus validation accuracy.

Inference: It was observed that Dropout 0.5 produced a small validation improvement in this run.

Kernel, filter, stride, padding, pooling and channel count describe how spatial features change through a CNN. Depthwise separable convolution performs a depthwise spatial operation followed by a pointwise 1×1 channel-mixing operation, which is why MobileNetV2 is computationally efficient.

11 Transfer Learning and Fine-Tuning

Strategy	Learning rate	Trainable	Best val. accuracy	Time (s)
Feature extraction	0.001	Classifier head	91.17%	0.58
Fine-tuning	10^{-5}	Head + final 2 blocks	91.44%	170.93

Feature extraction froze the full MobileNetV2 base and trained only the new classifier. Fine-tuning then enabled the final two MobileNetV2 feature blocks, kept all Batch Normalization layers frozen, and used a smaller learning rate. A smaller learning rate protects useful ImageNet features from being changed too quickly while the new task-specific features adapt.

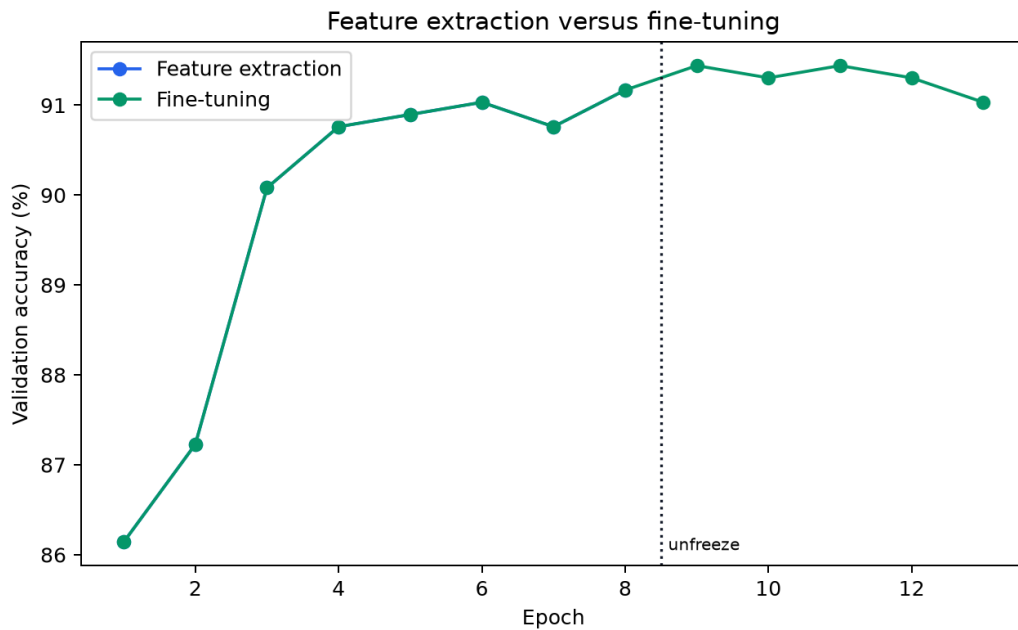


Figure 13: Validation accuracy before and after fine-tuning.

Inference: It was observed that validation accuracy improved after the upper MobileNetV2 blocks were unfrozen.

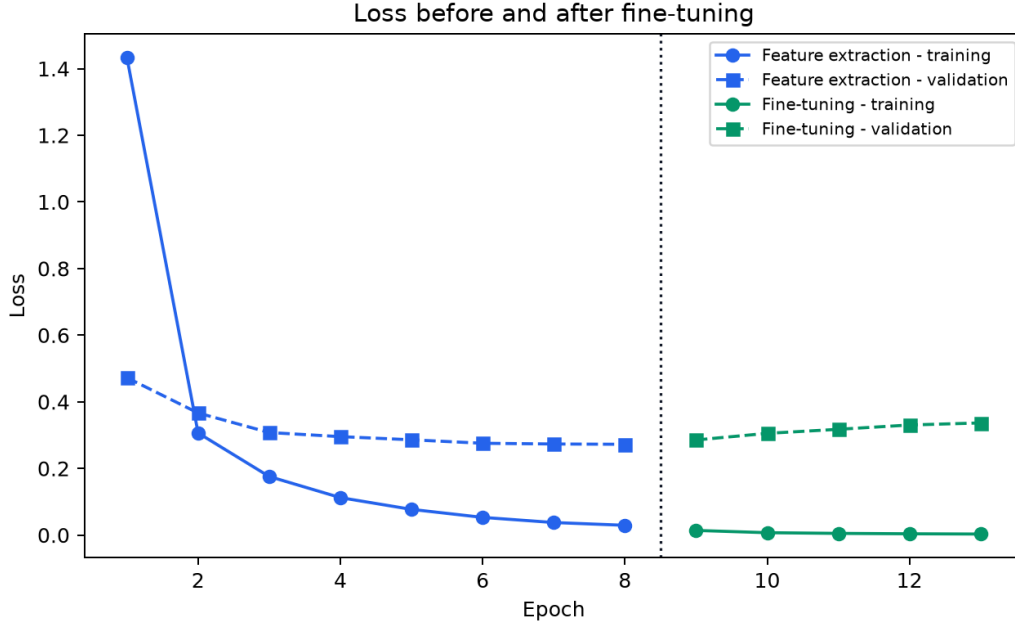


Figure 14: Training and validation loss before and after fine-tuning.

Inference: It was observed that the loss decreased further after fine-tuning while the smaller learning rate kept the update stable.

12 K-Fold Cross-Validation

Four promising frozen-base configurations were evaluated on the training portion only. Each fold trained a fresh 128-unit classifier on the cached pretrained features; the five validation accuracies, mean and population standard deviation are reported below.

Configuration	F1	F2	F3	F4	F5	Mean $\pm$ SD
C1 Baseline	92.19%	91.68%	91.51%	92.02%	90.65%	91.61 $\pm$ 0.54%
C2 L2	92.36%	91.51%	91.51%	91.68%	90.31%	91.47 $\pm$ 0.66%
C3 Dropout	92.02%	91.51%	91.68%	92.36%	90.31%	91.58 $\pm$ 0.70%
C4 BN + low LR	87.44%	85.23%	86.42%	84.55%	84.52%	85.63 $\pm$ 1.13%

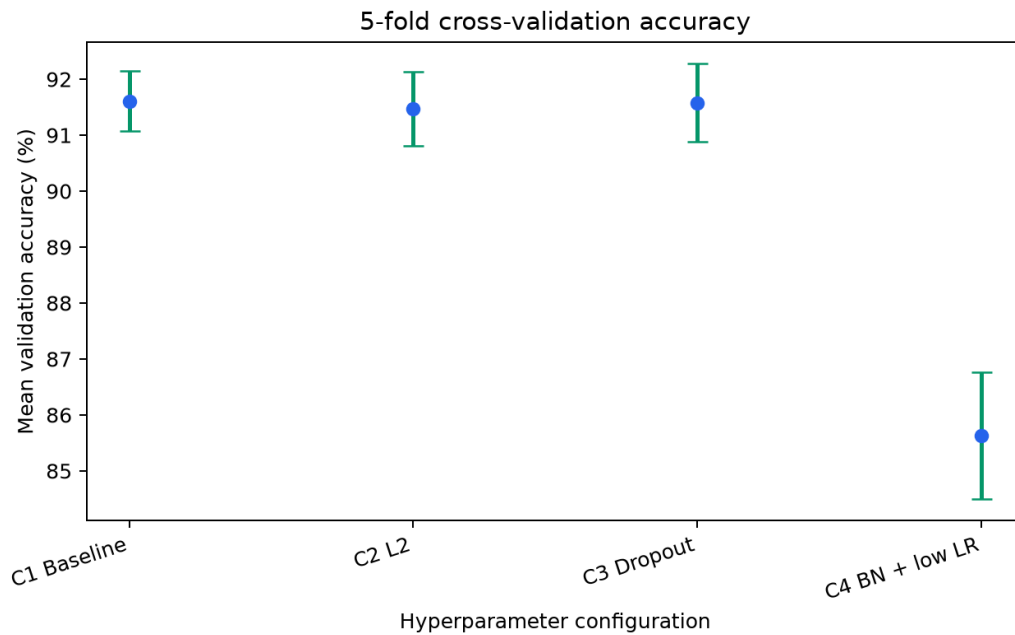


Figure 15: Mean 5-fold validation accuracy with standard-deviation error bars.

Inference: It was observed that C1 gave the strongest mean validation accuracy with a small fold-to-fold spread.

The selected configuration was C1 Baseline because it had the best mean validation accuracy after considering the fold-to-fold spread. Its mean was 91.61% with a standard deviation of 0.54 percentage points.

13 Final Model Evaluation

Metric	Value
Mean CV accuracy	91.61%
CV standard deviation	0.54 percentage points
Test accuracy	89.75%
Weighted precision	0.9015
Weighted recall	0.8975
Weighted F1-score	0.8972
Training time	1.29 s
Number of parameters	168,741 head parameters; 3,538,984 MobileNetV2 parameters

The selected CV configuration was retrained on all training images. The validation split was retained for model-development checks, and the official test set was evaluated only at this final stage. The fine-tuned MobileNetV2 result is shown separately because it is the transfer-learning comparison model.

Class	Precision	Recall	F1-score	Support
Abyssinian	0.8763	0.8673	0.8718	98
American Bulldog	0.7218	0.9600	0.8240	100
American Pit Bull Terrier	0.8356	0.6100	0.7052	100
Basset Hound	0.9368	0.8900	0.9128	100
Beagle	0.8545	0.9400	0.8952	100
Bengal	0.7627	0.9000	0.8257	100
Birman	0.7477	0.8300	0.7867	100
Bombay	0.8269	0.9773	0.8958	88
Boxer	0.8788	0.8788	0.8788	99
British Shorthair	0.8721	0.7500	0.8065	100
Chihuahua	0.8854	0.8500	0.8673	100
Egyptian Mau	0.8515	0.8866	0.8687	97
English Cocker Spaniel	0.9412	0.9600	0.9505	100
English Setter	0.9787	0.9200	0.9485	100
German Shorthaired	0.9091	1.0000	0.9524	100
Great Pyrenees	0.9798	0.9700	0.9749	100
Havanese	0.8879	0.9500	0.9179	100
Japanese Chin	0.9798	0.9700	0.9749	100
Keeshond	0.9900	1.0000	0.9950	99
Leonberger	1.0000	0.9700	0.9848	100
Maine Coon	0.8941	0.7600	0.8216	100
Miniature Pinscher	0.9565	0.8800	0.9167	100
Newfoundland	0.9700	0.9700	0.9700	100
Persian	0.9444	0.8500	0.8947	100
Pomeranian	0.9691	0.9400	0.9543	100
Pug	0.9894	0.9300	0.9588	100
Ragdoll	0.6522	0.7500	0.6977	100
Russian Blue	0.8571	0.7800	0.8168	100
Saint Bernard	0.9245	0.9800	0.9515	100
Samoyed	0.9519	0.9900	0.9706	100
Scottish Terrier	0.9796	0.9697	0.9746	99
Shiba Inu	0.9524	1.0000	0.9756	100
Siamese	0.9767	0.8400	0.9032	100
Sphynx	0.9394	0.9300	0.9347	100
Staffordshire Bull Terrier	0.7215	0.6404	0.6786	89

Class	Precision	Recall	F1-score	Support
Wheaten Terrier	0.9592	0.9400	0.9495	100
Yorkshire Terrier	0.9697	0.9600	0.9648	100
Weighted average	0.9015	0.8975	0.8972	3669

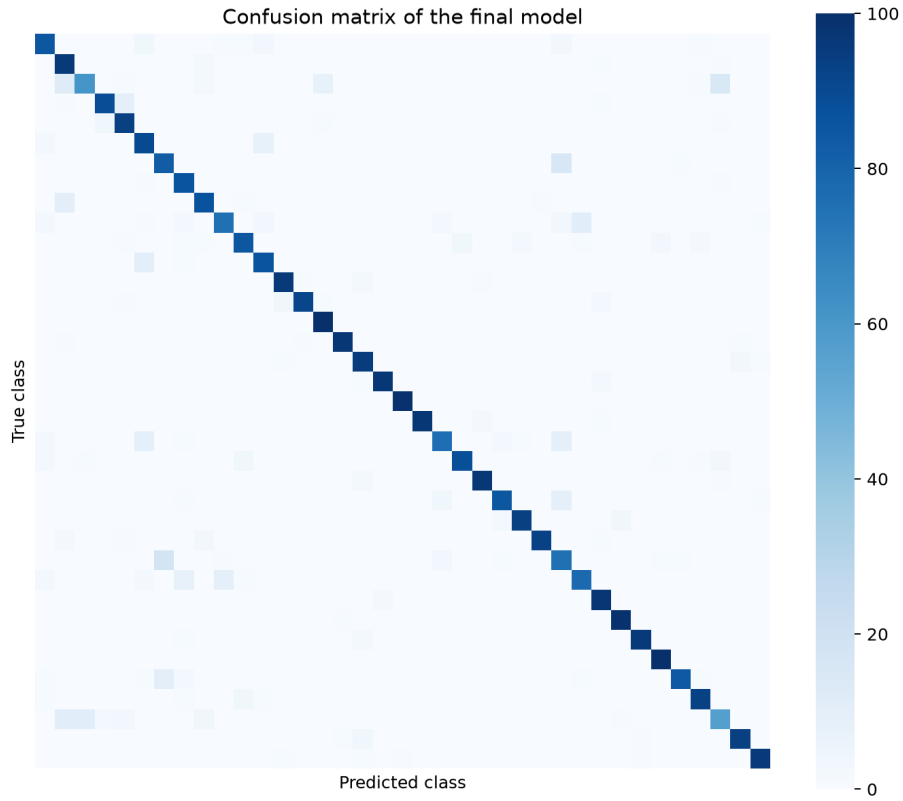


Figure 16: Confusion matrix of the final selected model.

Inference: It was observed that most predictions lay on the diagonal, while the largest confusion was Ragdoll predicted as Birman.

Representative misclassified test images

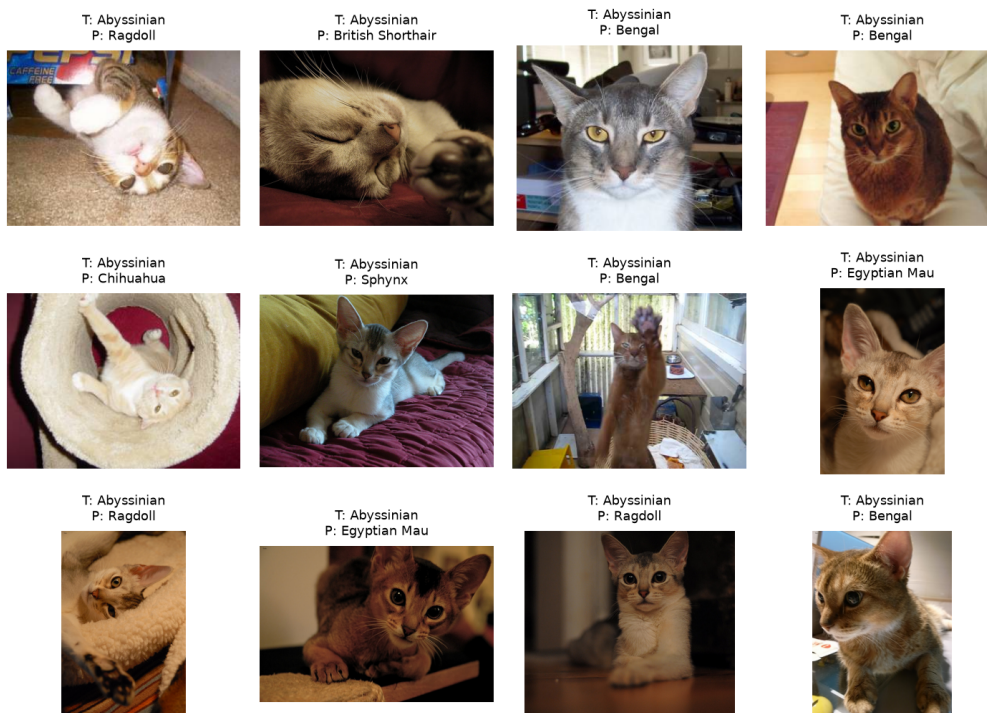


Figure 17: Representative misclassified test images.

Inference: It was observed that most errors occurred between breeds with similar colour, fur length or facial structure.

14 Overall Results

Configuration	CV accuracy	SD	Test accuracy	Training time	Parameters
Baseline frozen head	91.61%	0.54%	89.75%	2.56 s	168,741
Best initialization (Xavier/Glorot)	91.61%	0.54%	89.75%	1.29 s	168,741
Best regularization (L2)	91.24%	0.41%	89.04%	2.77 s	168,741
Best optimizer (RMSProp)	91.81%	0.49%	89.21%	1.85 s	168,741
Best hyperparameters	91.03%	0.35%	89.42%	1.89 s	168,741
Fine-tuned MobileNetV2	—	—	89.10%	170.93 s	2,392,613

The final choice balances validation accuracy, variability and computational cost. Fine-tuning improved the validation curve but did not beat the CV-selected frozen configuration on the held-out test set, while also requiring much more computation.

15 Discussion Questions

Q	Answer
1	Model parameters are learned weights and biases. Hyperparameters are choices set before training, such as learning rate, batch size and Dropout rate.
2	Good initialization keeps activations and gradients in a useful range, so learning starts smoothly and converges faster.
3	Zero initialization gives identical neurons identical gradients, so the neurons do not learn different features. In the experiment, the zero-initialized head stayed near chance.
4	Xavier initialization is suited to balanced input and output variance, while He initialization is designed for ReLU activations and usually gives a stronger start here.
5	Overfitting appears when training accuracy keeps rising while validation accuracy stops improving or validation loss rises.
6	Dropout randomly removes hidden activations during training, so the model cannot rely on one small set of features and usually generalizes better.
7	Batch Normalization stabilizes intermediate activations and provides learnable scale and shift parameters.

8	For [2, 4, 6, 8], the mean is 5, the variance is 5, and the normalized values are approximately [-1.342, -0.447, 0.447, 1.342].
9	Gamma controls the learned scale and beta controls the learned shift after normalization.
10	SGD uses the current gradient, Momentum accumulates direction, RMSProp adapts the step using recent squared gradients, and Adam combines momentum with adaptive scaling.
11	A learning rate that is too large can overshoot useful minima, produce unstable curves and even make the loss diverge.
12	A learning rate that is too small makes progress slow and can leave the model under-trained within the available epochs.
13	A larger batch gives smoother gradient estimates and fewer updates per epoch, but can reduce the number of useful parameter updates and use more memory.
14	Stride is the spatial step of a filter. Padding adds border values so the output can preserve more spatial positions.
15	MobileNetV2 replaces many full convolutions with depthwise separable convolutions and uses narrow bottlenecks, reducing parameters and computation.
16	Depthwise separable convolution applies spatial filtering channel by channel, then mixes channels with a 1×1 pointwise convolution.
17	Transfer learning reuses features learned from a large source dataset such as ImageNet for a new target task.
18	Feature extraction freezes the pretrained base. Fine-tuning unfreezes selected upper layers and adjusts them with a small learning rate.
19	A smaller rate reduces the chance of destroying useful pretrained features while the target-specific layers adapt.
20	5-fold cross-validation measures performance across several training and validation partitions, making model selection less dependent on one split.
21	Using the test set during tuning makes it part of the development process and gives an optimistic final estimate.
22	The mean describes expected validation performance, while the standard deviation describes how sensitive the result is to the split.
23	No. A slightly lower mean with much lower variability or much lower cost may be the better practical choice.

16 Additional Exercise

Two new combinations were evaluated with the same 5-fold training-only protocol. Both were compared with the selected configuration using mean accuracy, standard deviation and training cost.

Configuration	Learning rate	Dropout	Batch size	Fine-tuning	Mean $\pm$ SD	Test accuracy
C5 New: low LR + Dropout	10^{-4}	0.50	16	Frozen base	$87.30 \pm 1.11\%$	87.54%
C6 New: large batch + Dropout	10^{-3}	0.25	64	Frozen base	$91.27 \pm 0.77\%$	89.37%

The new combinations did not improve the selected configuration's mean validation accuracy enough to justify their extra variability or cost. The selected model was therefore preferable for this run because it was simpler and more stable on the training-only folds.

17 Expected Outcome and Conclusion

The experiments showed that initialization, regularization, optimizer choice and hyperparameters affected both convergence and generalization. MobileNetV2 supplied a compact pretrained representation; fine-tuning the upper blocks improved the target-task fit, while 5-fold cross-validation made the final frozen-base selection more reliable than a single validation split.

In this run, the CV-selected model reached 89.75% test accuracy with weighted F1-score 0.8972. The fine-tuned model reached 89.10% test accuracy; its validation improvement did not transfer to a higher held-out test score in this run.

18 References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.

2. Sergey Ioffe and Christian Szegedy, “Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift,” ICML, 2015.
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4. Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman and C. V. Jawahar, “Cats and Dogs,” CVPR, 2012.
5. Torchvision and PyTorch documentation used for the executable implementation.